

# Analyzing Relationship between VTubers and Their Highlight Editors in the Context of Online Collaboration

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## Abstract

This study examines the relationship between VTubers and their highlight editors as a case of collaborative dynamics in online content ecosystems. We hypothesize a reciprocal interaction in viewership between original VTuber videos and highlight clips—namely, that each might increase the visibility of the other. However, through quantitative analysis using time-series and regression methods, we find no consistent statistical evidence of such causality. This suggests that direct mutual promotion in terms of viewership does not reliably occur. Instead, VTubers and editors appear to collaborate asynchronously and in decentralized ways, complementing one another by appealing to distinct segments of the audience. Through this indirect coordination, they co-create visibility and value, forming a mutually complementary relationship. Despite the absence of formal agreements or communication, their interaction is mediated by platform affordances and recommendation algorithms. This model of collaboration exemplifies a spontaneous, unofficial form of cooperation that characterizes contemporary digital entertainment cultures.

## CCS Concepts

• **Human-centered computing** → **Computer supported cooperative work**.

## Keywords

participatory culture, VTuber, Model of Coordinated Action, MoCA, user-generated content, media ecology

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## 1 Introduction

How do people collaborate creatively over the internet? Existing media and communication theories have long emphasized that shared media enables distributed actors to align and collaborate—even in the absence of formal structure or consensus. Star and Griesemer's influential concept of *boundary objects* posits that heterogeneous communities can coordinate effectively by operating through shared representations or media artifacts [6]. Building on this premise, Jenkins theorized "Participatory Culture" as a fan-driven mode of engagement beyond passive consumption, and further developed "Spreadable Media" to emphasize user-led circulation and remix as central to contemporary cultural flow [2, 3]. These frameworks converge on a key assumption: shared media resources foster co-creation, coordination, and community engagement.

A particularly illustrative case of emergent, media-based collaboration is the VTuber ecosystem. In this space, full-length livestreams and user-generated highlight edits—commonly known as *kirinuki*—form a dynamic, interdependent media ecology. These actors (streamers and editors) operate informally yet in tandem: they share audio-visual assets, attract overlapping audience communities, and engage in mutual value creation through algorithmic visibility and monetization loops [1, 5]. Despite lacking formalized contracts or organizational boundaries, their actions are tightly coupled through shared media. This environment thus offers an ideal empirical context for examining how collaboration may occur under conditions of decentralization, structural asymmetry, and platform mediation. Recent large-scale work by Wei and Tyson [7] further demonstrates how VTuber fan communities shape engagement and monetization patterns on streaming platforms, underscoring the need for nuanced analysis of this collaborative media ecology.

To analyze this loosely coordinated but tightly linked relationship, we adopt the *Model of Coordinated Action* (MoCA) [4]. MoCA offers a multidimensional framework that captures coordination across seven axes—including synchronicity, turnover, communities of practice, and nascence—thus enabling analysis of emergent collaborative forms. We propose that the VTuber–kirinuki ecology exhibits many of these dimensions in pronounced ways: it is high-turnover, involves multiple (or even unaligned) communities of practice, and relies on an evolving, informally sustained coordination logic.

The broader aim of this study is to understand what kind of collaboration is taking place in this ecology. Specifically, we ask:

- **RQ1:** To what extent do VTubers and kirinuki editors influence each other through their media practices? Can we empirically identify mutual effects between original streams

and derivative highlight clips in terms of audience engagement and visibility?

- **RQ2:** What forms of coordination or collaboration emerge between VTubers and kirinuki editors, and how can these be characterized using the MoCA framework?

To address these questions, we combine quantitative time-series analysis (to examine mutual influence in viewership trends) with qualitative media ecosystem mapping (to characterize coordination patterns). While our empirical tests reveal no statistically significant bidirectional causality between clip and stream views, qualitative evidence suggests a persistent, interdependent relationship sustained by audience segmentation and algorithmic amplification.

This paper makes two core contributions:

- (1) **Empirical insight:** We demonstrate that the VTuber–kirinuki ecology functions as a form of distributed collaboration, with each actor occupying distinct but complementary roles in audience engagement.
- (2) **Theoretical extension:** By applying the Model of Coordinated Action (MoCA) to the VTuber–kirinuki ecosystem, we illustrate how MoCA can be operationalized to analyze coordination in platform-mediated environments involving emergent practices and informal actors.

## 2 Methods & Findings

To examine patterns of coordination within the VTuber–kirinuki ecosystem, we adopted a multi-method empirical strategy aimed at surfacing potential causal mechanisms and asymmetric interdependencies. Specifically, we applied a combination of correlation visualizations, Granger causality testing, ordinary least squares (OLS) regression, and supplementary time-series and cross-sectional analyses. Our central aim was to assess whether highlight clips significantly contribute to the growth of original VTuber channels, and to explore the relationship.

### Data Collection Overview

We collected structured metadata from Holodex, a platform aggregating VTuber-related content. Our dataset includes information on both full-length stream videos and user-generated highlight clips, along with channel-level and viewership metrics. View counts for both original videos and highlight clips were logged hourly from April 1 to May 10, 2025.

- Total VTuber channels: 3,180
- VTuber channels with at least one associated clip: 440
- Hourly view count records across both clips and streams between April 1 and May 10, 2025.

These data were organized using a relational schema comprising tables for channels, original\_videos, clips, and corresponding view\_counts. This schema supported scalable panel-based time-series and regression modeling.

### Viewership Growth and Clip Activity

We began by plotting initial full-stream view counts against subsequent growth, enriching each data point with clip-related metadata. In Figure 1, dot color represents the number of user-generated

highlight clips, while dot size corresponds to total cumulative view volume.

The scatterplot reveals a general trend: channels with high initial visibility and active clipping tend to grow disproportionately. However, several notable exceptions complicate this pattern. For instance, some highly viewed channels have relatively few associated clips. Many of these are official music or archive channels, where licensing constraints, the polished nature of content, or format-specific expectations make them less likely targets for user-generated clipping. These cases indicate that while clip activity plays a role in discoverability, its influence is mediated by content type and community norms.

### Causality Testing via Granger Analysis

To further investigate temporal dynamics, we conducted Granger causality tests using panel time-series data. Specifically, for each VTuber  $i$ , we tested whether past fluctuations in clip view counts significantly improved prediction of future subscriber counts beyond what could be predicted by past subscriber counts alone.

Formally, we estimated the following restricted and unrestricted vector autoregressive (VAR) models:

$$S_{it} = \alpha_i + \sum_{p=1}^P \beta_p S_{i,t-p} + \epsilon_{it}^{(R)} \quad (\text{restricted})$$

$$S_{it} = \alpha_i + \sum_{p=1}^P \beta_p S_{i,t-p} + \sum_{q=1}^Q \gamma_q C_{i,t-q} + \epsilon_{it}^{(U)} \quad (\text{unrestricted})$$

where:

- $S_{it}$  is the subscriber count for channel  $i$  at time  $t$
- $C_{i,t-q}$  is the lagged clip view count
- $P$  and  $Q$  are the lag lengths

The null hypothesis  $H_0$  is that all  $\gamma_q = 0$ , meaning that past clip activity does not Granger-cause subscriber growth. We computed the F-statistic:

$$F = \frac{(SSR_R - SSR_U)/Q}{SSR_U/(N - K)}$$

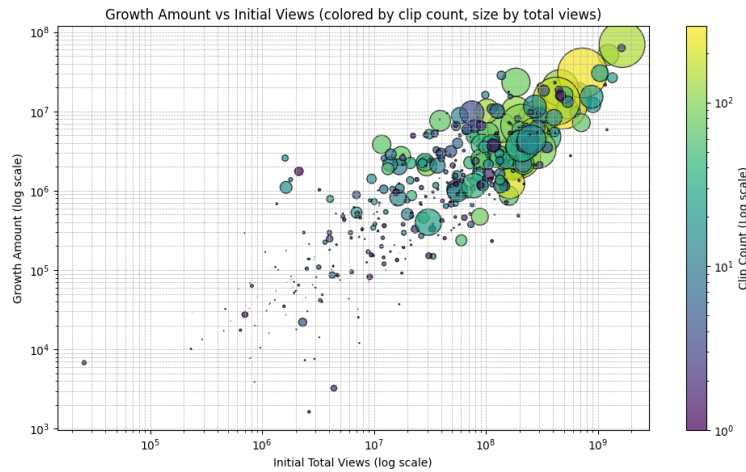
where  $SSR_R$  and  $SSR_U$  are the sum of squared residuals from the restricted and unrestricted models,  $Q$  is the number of lagged clip terms,  $N$  is the number of observations, and  $K$  is the total number of parameters in the unrestricted model.

Figure 2 visualizes channels with significant Granger causality ( $p < 0.05$ ). Only a subset of channels exhibited significant clip-driven effects, indicating that highlight editors' impact varies and is not uniformly distributed across the ecosystem.

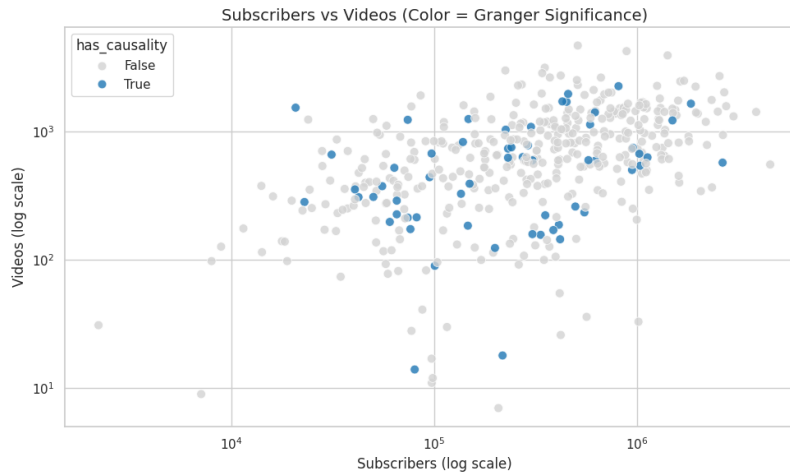
### OLS Regression: Predictive Power of Clip Views

We further examined the relationship between aggregate clip viewership and subscriber gain using OLS regression. Table 1 summarizes the model output.

Although the coefficient was statistically significant ( $p = 0.005$ ), the effect size was modest ( $R^2 = 0.001$ ), implying that clip activity alone offers limited predictive power for channel growth.



**Figure 1: Growth in viewership vs. initial views (log-log scale).** Dot color represents the number of highlight clips; dot size reflects total clip view volume. Brighter, larger circles suggest stronger clip-mediated viral growth.



**Figure 2: Subscribers vs. Videos for VTuber channels (both axes in log scale).** Dot color indicates Granger-significant causality from clips to channel growth; size reflects the number of user-generated highlight clips. Blue dots denote statistically significant causality.

**Table 1: OLS Regression: Total Clip Views → Subscriber Growth**

| Variable         | Coef.  | <i>p</i> -value |
|------------------|--------|-----------------|
| Intercept        | 10.67  | 0.000           |
| Total Clip Views | 0.0002 | 0.005           |

- We tested for bidirectional Granger causality between full-stream and clip view trajectories—results were not statistically significant.
- We examined interaction terms such as genre, clip length, editor diversity, and monetization status, but none robustly explained outcome variation.
- Cluster analyses based on content theme, streaming frequency, and viewer demographics did not yield stable typologies correlated with asymmetric growth.

### Additional Analyses and Null Results

Beyond the headline models, we explored several other statistical avenues:

These null findings suggest that while clips may facilitate visibility, they are not sufficient as standalone drivers of success, nor do they guarantee mutual benefit across actor roles.

## Synthesis of Findings in Relation to Research Questions

With respect to our research questions, we draw the following conclusions from the combined quantitative and qualitative evidence:

- **RQ1:** Our time-series analyses, including Granger causality testing and regression modeling, provide limited evidence for statistically significant, bidirectional influence between VTuber stream views and highlight clip views. While certain channels—particularly mid-tier or niche creators—exhibit clip-driven growth, such effects are neither consistent nor strong across the dataset. Thus, we find that media practices between VTubers and editors are interdependent in form, but not predictively coupled in function. Visibility gains appear to emerge more from asynchronous, algorithmically mediated dynamics than from direct reciprocal influence. Furthermore, this interdependence reveals a division of labor in audience engagement: although both parties work with the same underlying content, they appear to target distinct viewer segments. VTubers primarily engage real-time or core fans through long-form streams, whereas kirinuki editors cater to casual, time-constrained, or algorithmically exposed viewers via short-form highlights. This asymmetry suggests that collaboration does not imply identical function, but rather complementary roles aligned to different modalities of attention and discovery.
- **RQ2:** Despite the weak statistical coupling, qualitative patterns suggest a sustained coordination logic between VTubers and kirinuki editors. This relationship is best described through the MoCA framework as high-turnover, low-synchronicity, and multi-community in character. Editors and streamers operate in informal but complementary roles, engaging distinct audience segments and co-creating value through iterative media reuse. Their coordination, though emergent and decentralized, reflects a shared understanding of platform affordances, audience expectations, and mutually beneficial visibility loops.

These findings support our broader claim that the VTuber–kirinuki ecosystem constitutes a form of distributed collaboration, where structural asymmetry and informal alignment coexist within a platform-mediated media ecology.

## 3 Future Work

This study was based on a relatively short observation period; future research will examine whether longer-term data reveals different patterns of coordination and audience engagement over time. Analyzing longer time series will help uncover potential seasonal, cyclical, or cumulative effects in VTuber–kirinuki relationships, providing deeper insights into the sustainability of informal collaborations.

In addition to quantitative analyses, we plan to incorporate qualitative methods such as in-depth interviews with both VTubers and highlight editors. These perspectives will help contextualize our findings, illuminate platform-specific incentives, and unpack editorial strategies and audience targeting practices that may not be fully observable through metrics alone.

To better understand the platform-mediated nature of this ecosystem, we also intend to examine the role of recommendation algorithms, content discovery features, and monetization mechanisms. By analyzing how these platform-level factors shape coordination and audience segmentation, we can enrich our theoretical extension of the Model of Coordinated Action (MoCA) to account for algorithmic mediation.

Moreover, beyond the VTuber–kirinuki ecosystem, we intend to broaden the scope of inquiry to the wider domain of collaborative practices on the internet, especially within entertainment-oriented contexts. While successful coordination models such as open-source software development have been extensively studied in technical domains, their applicability to participatory entertainment production remains underexplored.

In particular, we are interested in whether similar models of distributed co-creation can be applied to narrative and storytelling media. Can entertainment fans not only remix but also co-author evolving storyworlds in ways that mirror the collaborative structures of open-source communities? To explore this question, we plan to design and deploy a prototype platform that enables users to collaboratively build fictional narratives through shared prompts, branching storylines, and modular contributions.

Through this expanded multi-method approach and exploration of related contexts, we aim to advance CSCW’s understanding of emergent collaborative labor, platform governance, and the future of participatory media.

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